Lab 7

By

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“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to--except as documented in a bibliography or acknowledged below.

Signature /Caden Burnett/

*You may sign and scan the cover page, add a signature after converting to PDF, or type your name like this: */Joe Aggie/*

Artificial Intelligence (AI) Acknowledgement: Used Chat GPT to do a lot of writing after instructing it on what to say and how to structure things, and with what information to use.

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Contents

List of Figures 1
List of Tables 2
Executive Summary 3
Introduction 4
Methods 4
Results 6
Phase 1 6
Phase 2 7
Check Sheet and Pareto Diagram 8
Total Parts Numerical Comparison 12
Total Parts Graphical Analysis and Comparison 13
Good Parts Numerical Comparison 16
Good Parts Graphical Analysis and Comparison 17
Off-spec Parts Numerical Comparison 20
Off-spec Parts Graphical Analysis and Comparison 21
Bad Parts Numerical Comparison 25
Bad Parts Graphical Analysis and Comparison 26
Conclusions 29
Recommendations 30

List of Figures

Figure 1. X-bar Control Chart of Button-Hole Diameter 7
Figure 2. Time Series Plot of Subgroup value for Button-Hole Diameter 8
Figure 3. Pareto Chart for Phase 2 Nonconformities 12
Figure 4. Run Chart for Phase 2 Total Parts 13
Figure 5. Histogram of Phase 2 Total Parts 14
Figure 6. Probability Plot for Phase 2 Total Parts 15
Figure 7. Box Plot for Phase 2 Total Parts 16
Figure 8. Run Chart for Phase 2 Good Parts 17

Figure 9. Histogram of Phase 2 Good Parts 18
Figure 10. Probability Plot of Phase 2 Good Parts 19
Figure 11. Box Plot for Phase 2 Good Parts 20
Figure 12. Run Chart for Phase 2 Off-spec Parts 22
Figure 13. Histogram of Phase 2 Off-spec Parts 23
Figure 14. Probability Plot of Phase 2 Off-spec Parts 24
Figure 15. Box Plot of Phase 2 Off-spec Parts 25
Figure 16. Run Chart of Phase 2 Bad Parts 26
Figure 17. Histogram of Phase 2 Bad Parts 27
Figure 18. Probability Plot of Phase 2 Bad Parts 28
Figure 19. Box Plot of Phase 2 Bad Parts 29

List of Tables

Table 1. Phase 1 Basic Statistics 6
Table 2. Check Sheet for Phase 2 Nonconformities 9
Table 3. Basic Statistics for Phase 2 Total Parts 12
Table 4. Basic Statistics for Phase 2 Good Parts 16
Table 5. Basic Statistics for Phase 2 Off-spec Parts 21
Table 6. Basic Statistics for Phase 2 Bad Parts 26

1. Executive Summary

This report evaluates how the implementation of a control chart for Button Hole Diameter influenced production quality and process performance during Phase 2 of the Mouse Factory SPC simulation. The analysis shows that good part output increased significantly compared to Lab 5 data, with the Phase 2 mean rising from 591.45 to 616.45, and the t-test confirming that this improvement was statistically significant. Bad part counts declined substantially, with the average falling from 151.20 to 123.60, and a highly significant p-value indicating that the reduction in defects was unlikely to be due to random variation. Off-spec and total parts did not exhibit statistically meaningful changes, as their confidence intervals overlapped with the Lab 5 results, and t-tests produced p-values well above the significance threshold. These results indicate that the introduction of the control chart improved the consistency and quality of production even though it did not alter overall output volume. Continued use of the control chart is recommended along with expanded monitoring of additional process variables so that remaining sources of variation can be identified and addressed more effectively.

2. Introduction

The Phase I portion of this project involved performing a retrospective analysis using the Mouse Factory web-based SPC simulation tool. The objective of this phase was to develop statistically valid control limits for the Button-Hole Diameter characteristic, which was previously identified in Lab 5 as the process variable with the highest frequency of nonconforming occurrences. To ensure that the control chart is based on stable process behavior, the retrospective mode uses a special quality program that forces the process to operate under conditions of statistical control. Only common-cause variability is present during data collection, and no special-cause signals are introduced. This controlled environment allows for unbiased estimation of the process mean and standard deviation.

Phase II uses these established control limits to evaluate real-time production data and assess whether the implementation of SPC meaningfully affected process performance. By comparing Phase II output to the baseline results from Lab 5, this phase makes it possible to determine whether the control chart improved process stability, increased the number of conforming units, or reduced defect-producing variation. In this way, Phase II serves as a benchmark analysis against prior system performance, allowing for a direct evaluation of the practical impact of statistical process control on the factory's quality outcomes.

3. Methods

A rational subgroup size of n = 10 was selected to capture short-term variability in the Button-Hole Diameter measurements. The simulation tool was used to collect 15 subgroups, resulting in a total of 150 observations. The time to collect a subgroup in seconds was calculated to be 1,920 using Equation 1

$$Sampling\ Interval \leq \frac{total\ number\ of\ seconds\ in\ a\ work\ day}{m} \tag{1}$$

In this equation, m is representative of the number of subgroups to collect.

The sampling method was set to Random, which distributes the ten observations for each subgroup evenly across the sampling interval and produces more representative within-subgroup variation. For each subgroup, the individual measurements were automatically recorded, and the corresponding subgroup means and ranges were computed.

The retrospective dataset was then used to construct the initial X̄-chart. Because the retrospective analysis is performed under artificially stable operating conditions, any resulting control limits reflect true baseline process behavior and are appropriate for use during Phase II online monitoring.

After establishing the control limits during the retrospective analysis, the X̄ chart for the Button-Hole Diameter characteristic was implemented in the online monitoring mode of the Mouse Factory simulation. The control limits calculated in Phase I were entered into the online monitoring interface, and the system was run for 20 days. During this period, the numbers of total parts, good parts, bad parts, and off-spec parts produced each day were recorded. These daily counts were used to create benchmark metrics that quantify plant performance with the control chart deployed.

A numerical and graphical analysis of the resulting data, including basic statistics, a run chart, histogram, probability plot, and box plot, was evaluated and compared to previous lab data. To provide further insight into the analysis of the control chart data, a confidence interval was calculated for total, good, off-spec, and bad parts using Equation 2.

$$\bar{X} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \tag{2}$$

$\bar{X}$ is the sample mean across 20 days, t is the critical value from the t-distribution at a confidence level of 95% (α=0.05), s is the sample standard deviation, and n is the sample size.

To evaluate the statistical significance of the difference in results between the control chart data from Phase 2 and the previous data, a one-sample t-test was calculated for total, good, off-spec, and bad parts using Equation 3.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \tag{3}$$

In this equation, t is the test statistic to be evaluated, $\bar{X}_1, \bar{X}_2$ are the sample means of the previous and control chart sample means, and S_1, S_2 are the respective sample standard deviations. After computing the t-statistic, the value is compared to the t-distribution to obtain a p-value. The p-value represents the probability of observing a sample difference at

least as extreme as the one measured if the true means were actually equal. A significance level of $\alpha = 0.05$was used; if $p < 0.05$, the null hypothesis was rejected and the difference in means was considered statistically significant. If $p > 0.05$, the null hypothesis was not rejected, indicating that the observed difference was not statistically meaningful.

Results

3.1. Phase 1

Table 1 presents the descriptive statistics for the button-hole diameter measurements collected during Phase 1.

Table 1. Phase 1 Basic Statistics

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median
Button-Hole Diameter	150	0	0.241405	0.0010262	0.0125687	0.20973	0.23314	0.241815

Variable	Q3	Maximum
Button-Hole Diameter	0.249025	0.26796

The sample mean diameter is 0.241405 inches, and the standard deviation is 0.0125687 inches, indicating relatively low variation in the process. Based on these values and the sample size of 150, the 95% confidence interval for the true mean button-hole diameter is (0.2394, 0.2434). This narrow interval shows that the process average is estimated with high precision and that day-to-day variation in diameter is minimal relative to the overall mean.

Figure 1 is the X-bar control chart for Button-Hole Diameter measurements.

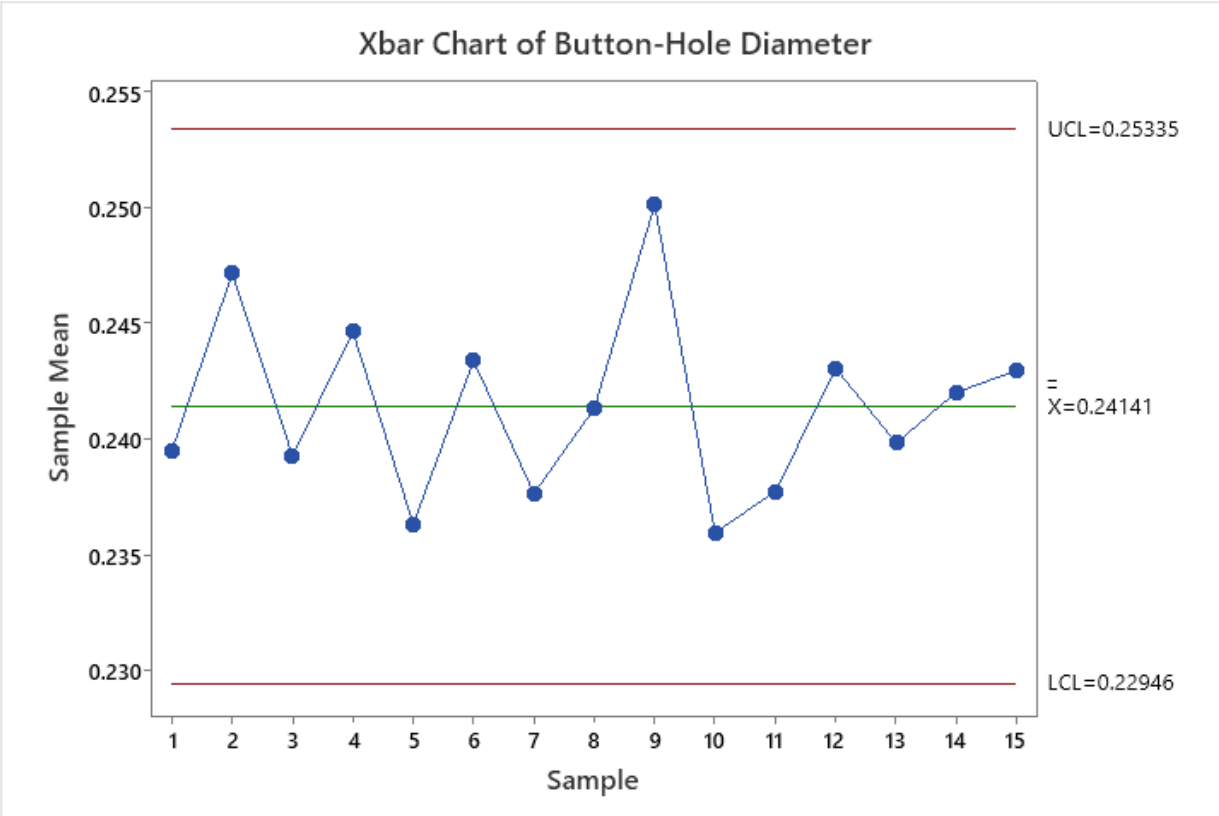

Figure 1. X-bar Control Chart of Button-Hole Diameter

This chart indicates that the Phase I data were in statistical control, as all subgroup means fall within the control limits (UCL = 0.25335, LCL = 0.22946) and no non-random patterns, trends, or shifts are present.

3.2. Phase 2

A time series plot in Figure 2 with the Phase 1 control limits overlaid was created to evaluate how the Button-Hole process behaved during the 20-day online monitoring period in Phase 2.

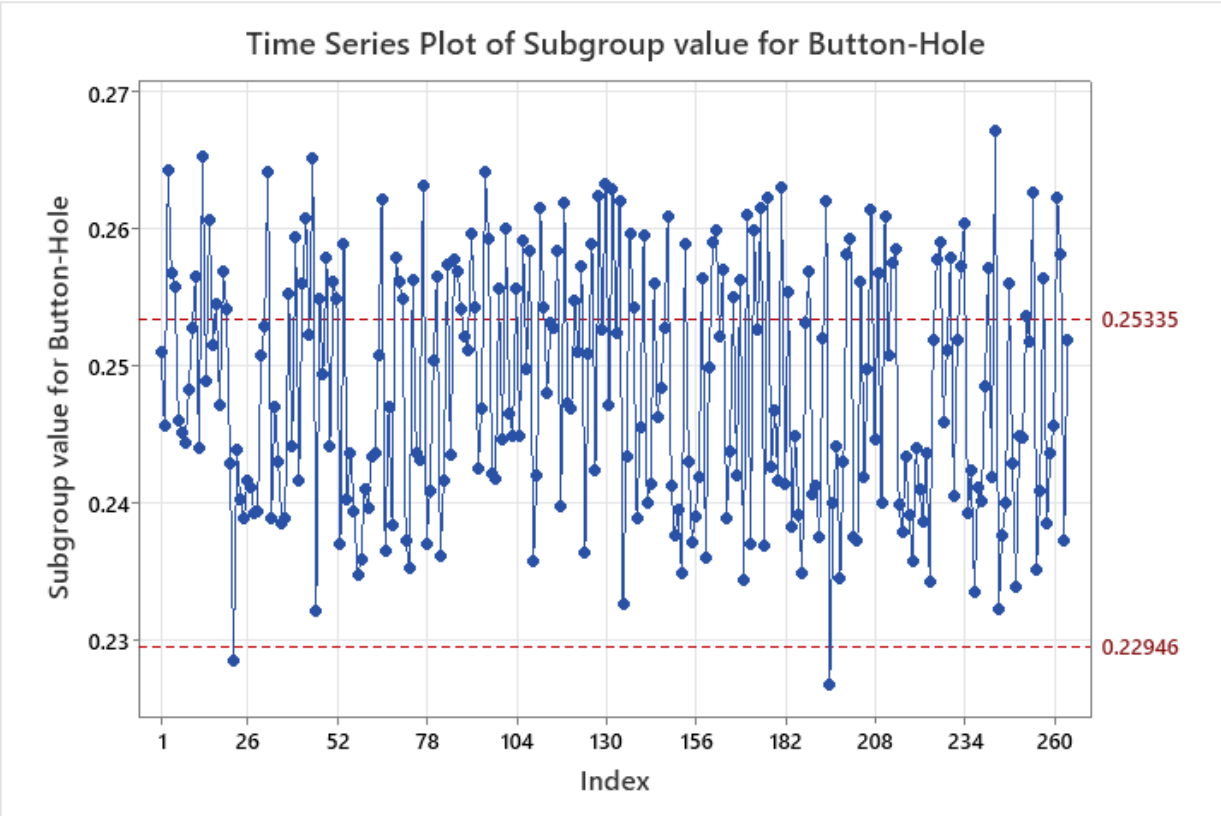

Figure 2. Time Series Plot of Subgroup value for Button-Hole Diameter

When the Phase 1 control limits (UCL = 0.25335, LCL = 0.22946) are added to the Phase 2 time series plot, it becomes clear that a large number of subgroup means exceed the upper control limit. The frequency and consistency of these violations indicate that the process did not maintain the statistical stability observed in Phase 1. Instead, the Button-Hole characteristic experienced a sustained upward shift in its operating level, with many points clustering above the UCL and very few approaching the LCL. This pattern suggests that assignable causes were present during online production and that the implemented control chart repeatedly identified true out-of-control conditions rather than occasional random variation. Overall, the Phase 2 time series demonstrates meaningful process deterioration relative to the Phase 1 baseline.

3.2.1. Check Sheet and Pareto Diagram

Using the quality data from phase 2 of the Mouse Factory data center, a check sheet (Table 2) and a corresponding Pareto chart were created to analyze the weight of certain defects amongst the population of both bad and off-spec parts, shown in Figure 3.

Table 2. Check Sheet for Phase 2 Nonconformities

Nonconformity	Week 1	Week 2	Week 3	Week 4	Grand Tota
Base-Black Streaking	4	4	4	4	16
Base-Bubble	4	8	3	2	17
Base-Burn Spot	2	5	3	3	13
Base-Mold Flashing	4	3	5	2	14
Base-Part Stuck in Mold	108	74	73	66	321
Base-Poor Weld Line	5	7	5	5	22
Base-Short Shot	2	2	1	4	9
Base-Sink Spot	2	7	5	2	16
Base-Snap Point Diameter	16	6	9	5	36
Base-Snap Point Opening	6	11	9	16	42
Base-Surface Imperfection	8	5	1	1	15
Base-Warpage	5	5	2	1	13
Button-Black Streaking	2	5	3	2	12
Button-Bubble	4	2	4	2	12
Button-Burn Spot	7	4	1	3	15
Button-Hole Diameter	108	155	110	107	480
Button-Mold Flashing	76	79	84	83	322
Button-Part Stuck in Mold	5	4	1	5	15
Button-Poor Weld Line	5	7	6	3	21
Button-Short Shot	5	4	5	1	15
Button-Sink Spot	5	4	4	5	18
Button-Surface Imperfection	4		1	4	9
Button-Warpage	8	9	2	3	22
Cover-Black Streaking	4	9	5	2	20
Cover-Bubble	6	6	6	6	24
Cover-Burn Spot	1	2	5	2	10
Cover-Mold Flashing	5	5	2	4	16
Cover-Part Stuck in Mold	5	3	5	3	16
Cover-Poor Weld Line	3	1	4	4	12
Cover-Shaft Diameter	13	11	11	8	43
Cover-Short Shot	3	3	2	4	12
Cover-Sink Spot	4	2	3	2	11
Cover-Surface Imperfection	4	4	1	5	14
Cover-Warpage	5	3	3	3	14
Diffuser-Black Specks	5	1	5	1	12
Diffuser-Brittleness	6	4	2	3	15
Diffuser-Burn Marks	3	6		4	13
Diffuser-Discoloration	3	4	2	5	14
Diffuser-Intensity	11	9	16	10	46
Diffuser-Jetting	3	3	4		10
Diffuser-Part Stuck in Mold	202	212	207	221	842
Diffuser-Splay Marks	2	1	1	3	7
Diffuser-Voids	3	1	4	1	9
Diffuser-Warpage	2	5	2	4	13

Diffuser-Weld lines	3	8	3	2	16
Scroll Base-Black Streaking	3	2		4	9
Scroll Base-Bubble	6	4	3	4	17
Scroll Base-Burn Spot	2	1	4	3	10
Scroll Base-Hole Diameter	11	16	11	7	45
Scroll Base-Mold Flashing	4	3	5	5	17
Scroll Base-Opening	14	14	7	12	47
Scroll Base-Part Stuck in Mold	1	3	2	3	9
Scroll Base-Pin Diameter	8	6	14	11	39
Scroll Base-Poor Weld Line	2	3	6	4	15
Scroll Base-Short Shot	4	3	4	3	14
Scroll Base-Sink Spot	4	4	2	5	15
Scroll Base-Surface Imperfecti	3	4	4	2	13
Scroll Base-Warpage	3	5	4	1	13
Scroll Wheel Assembly-Black S	8	1	5	5	19
Scroll Wheel Assembly-Bubble	8	4	4	4	20
Scroll Wheel Assembly-Burn S	5	3	1	2	11
Scroll Wheel Assembly-Missing	222	186	195	188	791
Scroll Wheel Assembly-Mold Fl	4	1	2	2	9
Scroll Wheel Assembly-Part St	5	2	2	4	13
Scroll Wheel Assembly-Shaft F	78	97	116	85	376
Scroll Wheel Assembly-Short S	1	4	3	6	14
Scroll Wheel Assembly-Sink Sp	2	7	4	1	14
Scroll Wheel Assembly-Surfac	5	6	4	3	18
Scroll Wheel Assembly-Warpa	2	5	4	4	15
Grand Total	1096	1097	1040	994	4227

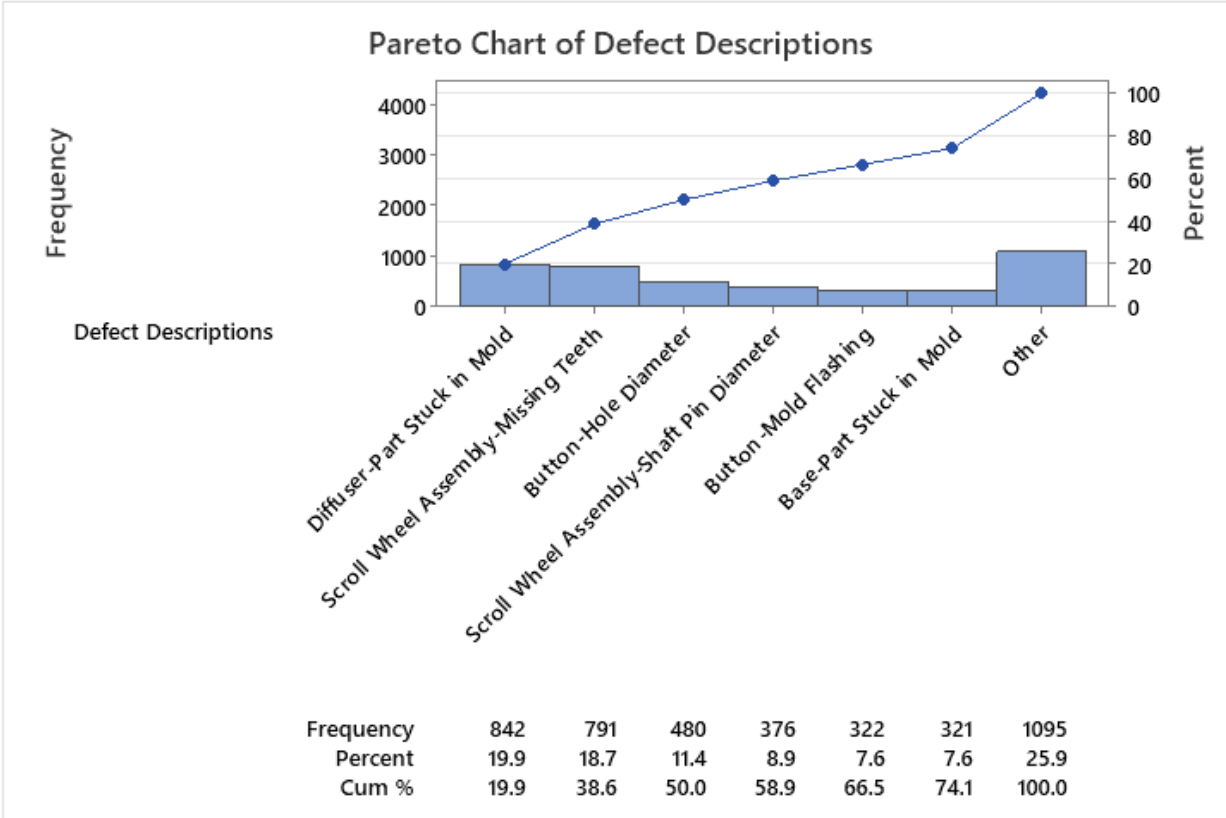

Figure 3. Pareto Chart for Phase 2 Nonconformities

The grand totals in the check sheet decreased from the previous data. Specifically, the Button-Hole diameter frequency decreased from 1072 to 480 according to the Pareto diagram.

3.2.2. Total Parts Numerical Comparison

Table 2 shows basic descriptive statistics for the Total Parts produced according to Phase 2 data.

Table 3. Basic Statistics for Phase 2 Total Parts

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Total	20	0	801.75	4.51190	20.1778	762	788.25	802	817.75	843

The mean is 801.75, which is slightly lower than the previous mean of 806.45. The standard deviation is 20.1778, which is greater than the previous standard deviation of 16.75. The 95% confidence interval is (792.31, 811.19), which is slightly wider and shifted down compared to the previous interval of (798.61, 814.29).

A one-sample t-test produces a p-value of around 0.428, indicating that there is no statistical difference between the control chart data with the previous data.

3.2.3. Total Parts Graphical Analysis and Comparison

A run chart of total daily output in Figure 4 was generated to evaluate whether the production process in Phase 2 exhibited random variation or signs of non-random behavior over time.

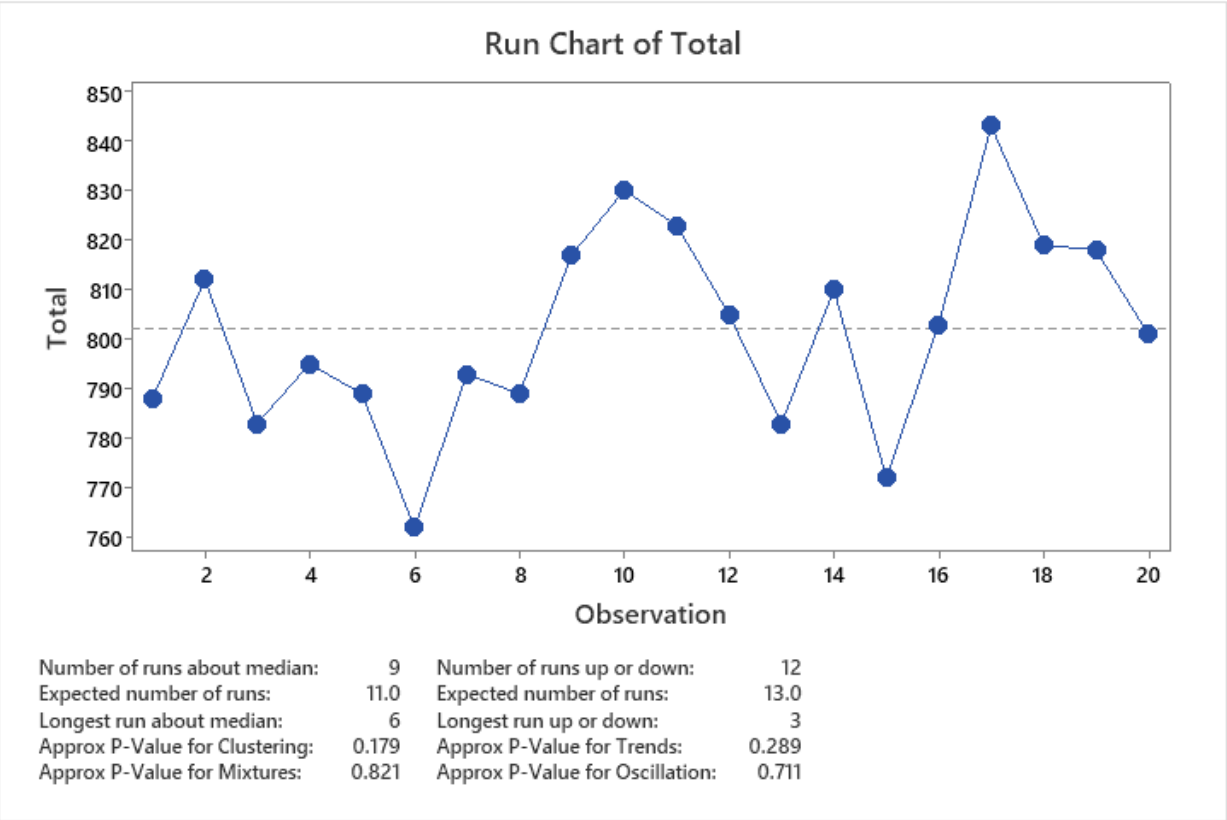

Figure 4. Run Chart for Phase 2 Total Parts

The chart shows natural fluctuation around the median, with the number of runs, run lengths, and corresponding p-values all falling within expected ranges. No evidence of clustering, trends, mixtures, or oscillation appears, indicating that total output varied randomly around its central level and behaved as a stable process throughout the period. Compared to the previous run chart, the earlier distribution displayed a much higher number of runs and a significant mixture p-value, suggesting more alternating behavior around the median and a less random pattern than what is observed in Phase 2.

A histogram of the total number of parts produced per day during the Phase 2 online monitoring period is shown in Figure 5.

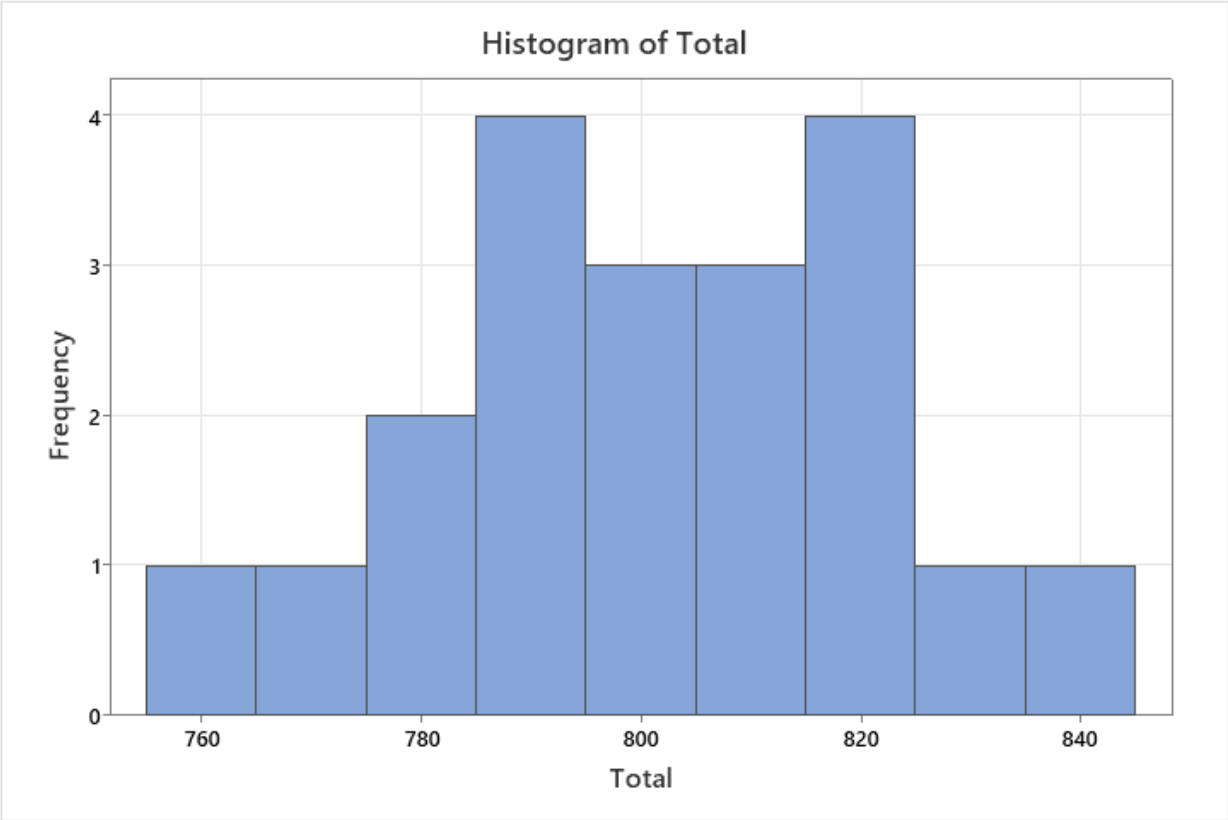

Figure 5. Histogram of Phase 2 Total Parts

The histogram shows that daily production is consistent, with most days centered around roughly 790–820 total parts. The spread is moderate, and no extremely low- or high-production days appear, indicating that the process maintained a steady throughput during the monitoring period. Compared to the earlier dataset, the Phase 2 totals show a similarly stable pattern of daily output, with no major changes in overall production volume.

A normal probability plot in Figure 6 of total daily parts was generated to assess whether the Phase 2 production totals follow a normal distribution.

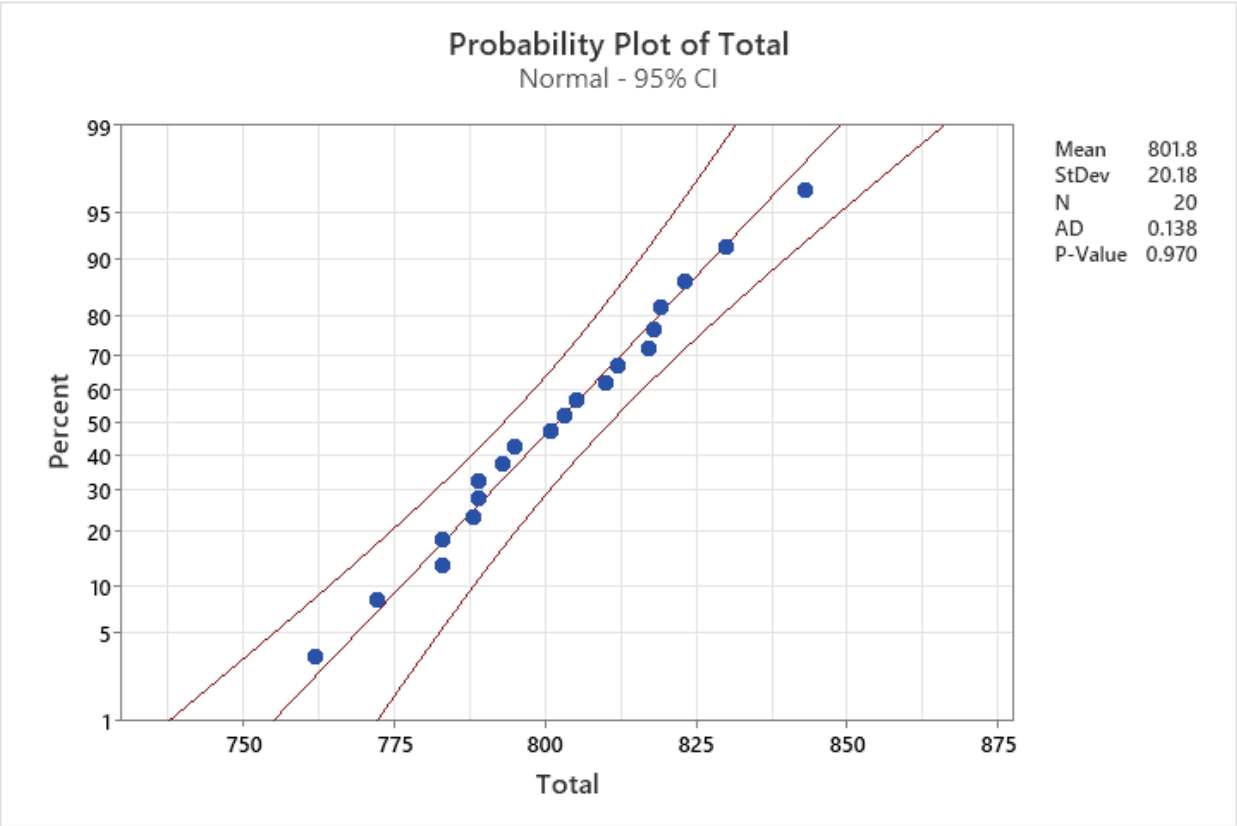

Figure 6. Probability Plot for Phase 2 Total Parts

The points align closely with the reference line and remain well within the confidence bands, indicating no meaningful deviations from normality. The high p-value (0.970) and low Anderson–Darling statistic support the conclusion that the daily total parts are consistent with a normal distribution. This normal behavior confirms that day-to-day production levels vary in a predictable, stable pattern suitable for confidence interval estimation and statistical comparison. Compared to the previous probability plot, the Phase 2 results show a similarly normal pattern, with no additional deviations or irregularities in the distribution of total daily production.

A boxplot of total daily parts in Figure 7 was created to summarize the central tendency and spread of production volume during Phase 2.

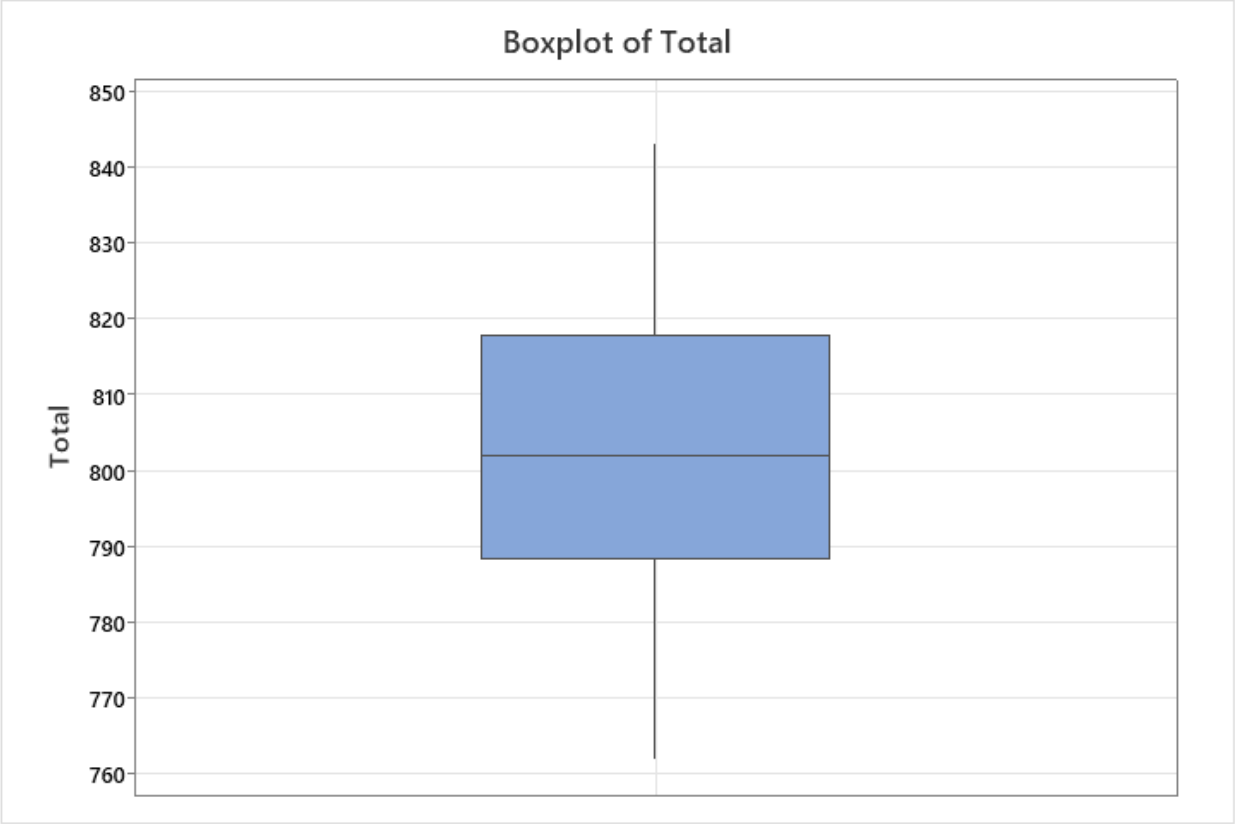

Figure 7. Box Plot for Phase 2 Total Parts

The boxplot shows that daily output is centered around roughly 800 parts, with moderate variability and no apparent outliers. The interquartile range is relatively narrow, indicating consistent production from day to day. The whiskers extend to reasonable bounds, suggesting that even the lowest and highest production days remained within an expected operating range. Compared to the previous boxplot, the Phase 2 distribution shows a similar central tendency and spread, indicating that overall production volume remained consistent across both periods.

3.2.4. Good Parts Numerical Comparison

Table 4 shows basic descriptive statistics for the Good Parts produced according to Phase 2 data.

Table 4. Basic Statistics for Phase 2 Good Parts

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Good Parts	20	0	616.45	3.75182	16.7787	591	600	614.5	629.75	646

The mean is 616.45, which is larger than the previous mean of 591.45. The standard deviation is 16.7787 which is slightly lower than the previous standard deviation of 16.83. The 95% confidence interval is (607.3, 625.6) which is slightly wider but shifted higher than the previous interval of (583.57, 599.33).

A one-sample t-test produces a p-value less than 0.001, indicating that the difference is statistically significant.

3.2.5. Good Parts Graphical Analysis and Comparison

A run chart of daily good parts in Figure 8 was created to assess whether in-spec production exhibited random variation or any signs of systematic behavior during Phase 2.

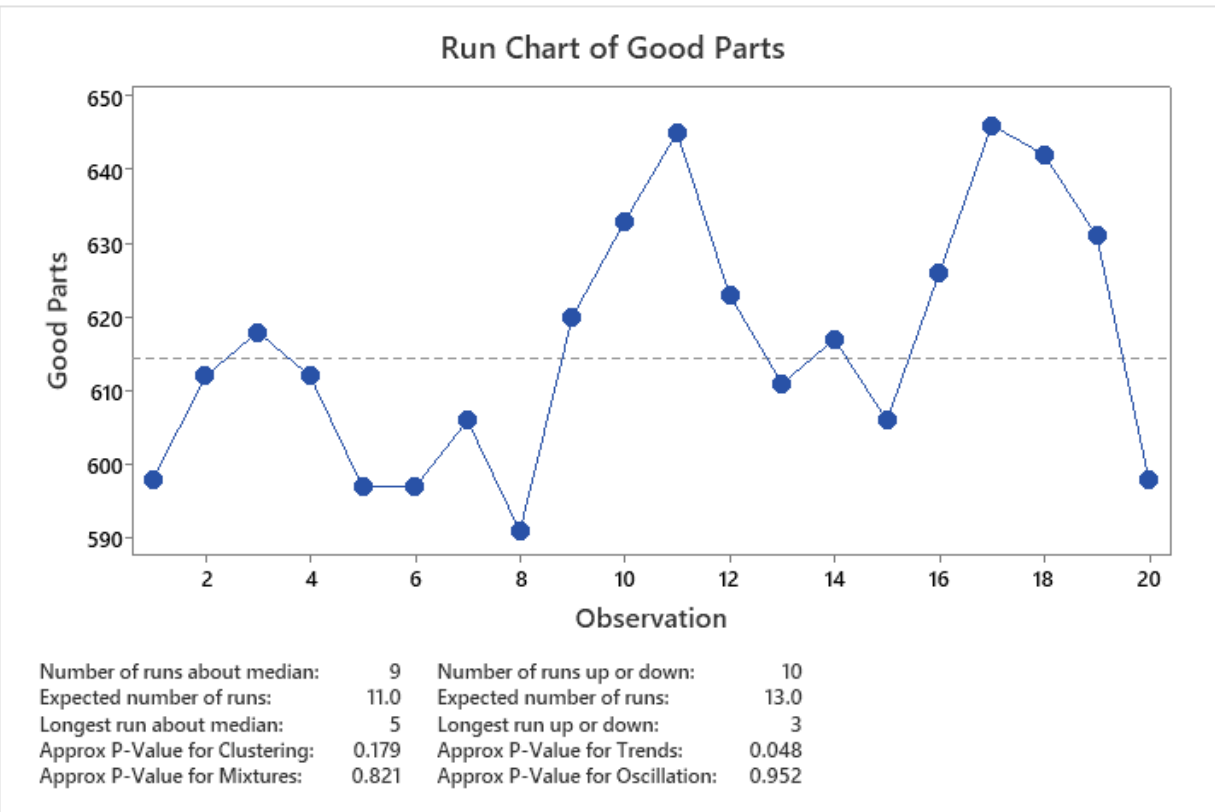

Figure 8. Run Chart for Phase 2 Good Parts

The chart shows that good-part output varies randomly around the median, with run counts and p-values indicating no evidence of clustering, mixtures, trends, or oscillation. This suggests that Phase 2 production operated with routine variation and without systematic patterns. Compared to the previous run chart, the Phase 2 pattern appears similarly stable, with no unusual run behavior or nonrandom signals detected in either period.

A histogram of daily good parts in Figure 9 was created to summarize how consistently the process produced in-spec units during Phase 2.

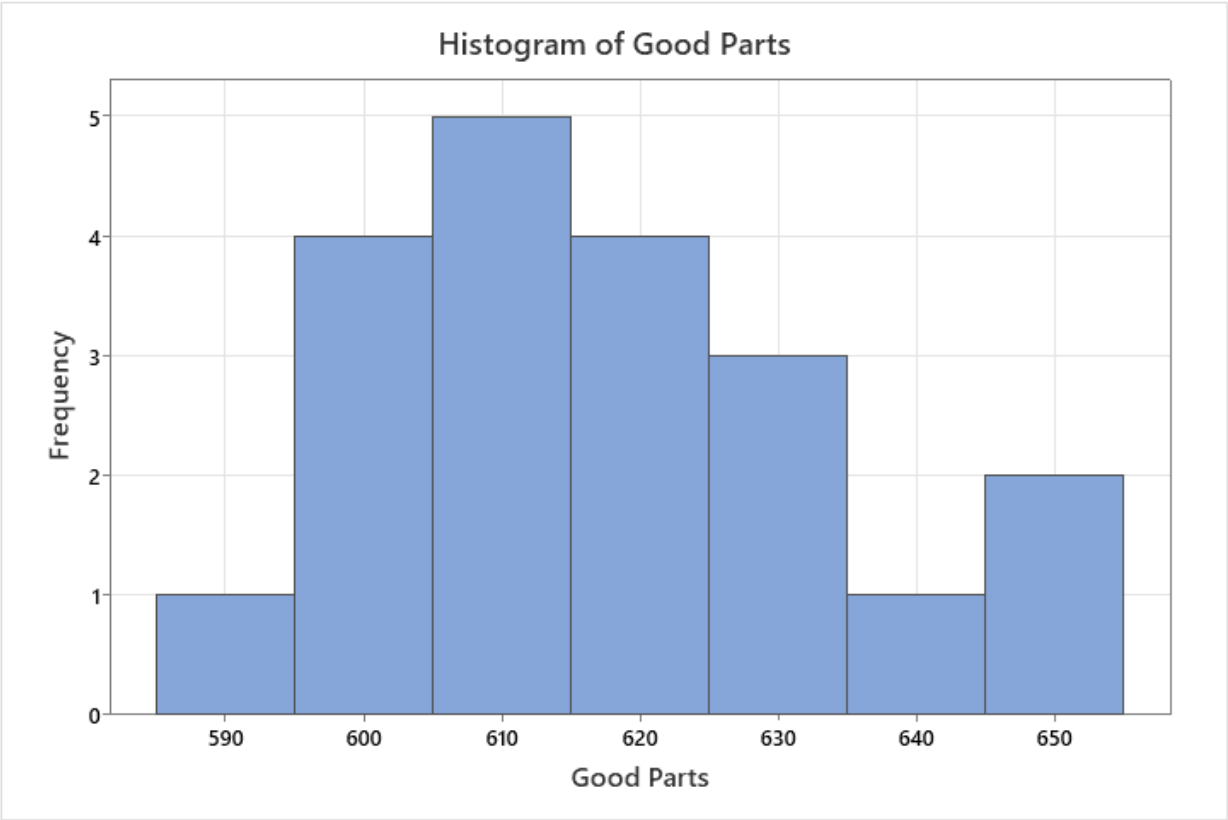

Figure 9. Histogram of Phase 2 Good Parts

The distribution centers around roughly 585–600 good parts, with several days extending into the higher-output range. This indicates generally stable in-spec production with more frequent higher-count days than before. Compared to the previous histogram, the Phase 2 results show an overall increase in good-part output.

A normal probability plot of daily good parts in Figure 10 was generated to evaluate whether in-spec production follows a normal distribution during Phase 2.

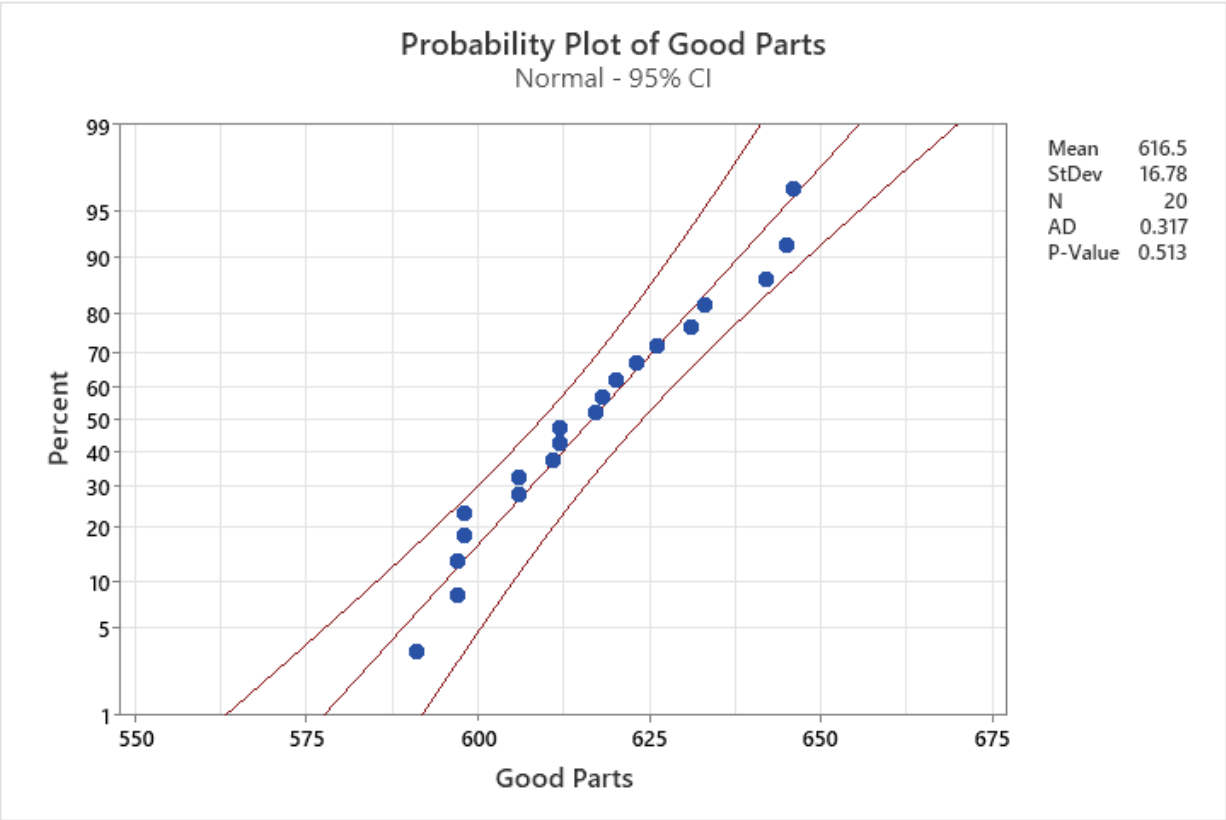

Figure 10. Probability Plot of Phase 2 Good Parts

A normal probability plot of daily good parts was generated to assess whether in-spec production follows a normal distribution during Phase 2. The points closely follow the reference line and remain within the confidence bands, indicating no meaningful departures from normality, and the high p-value (0.513) supports the conclusion that good-part output varies in a stable, predictable pattern. Compared to the previous probability plot, the Phase 2 results show a similarly strong fit to normality, indicating that the distributional behavior of good-part production remained consistent across periods.

A boxplot of daily good parts in Figure 11 was created to summarize the central tendency and variability of in-spec production during Phase 2.

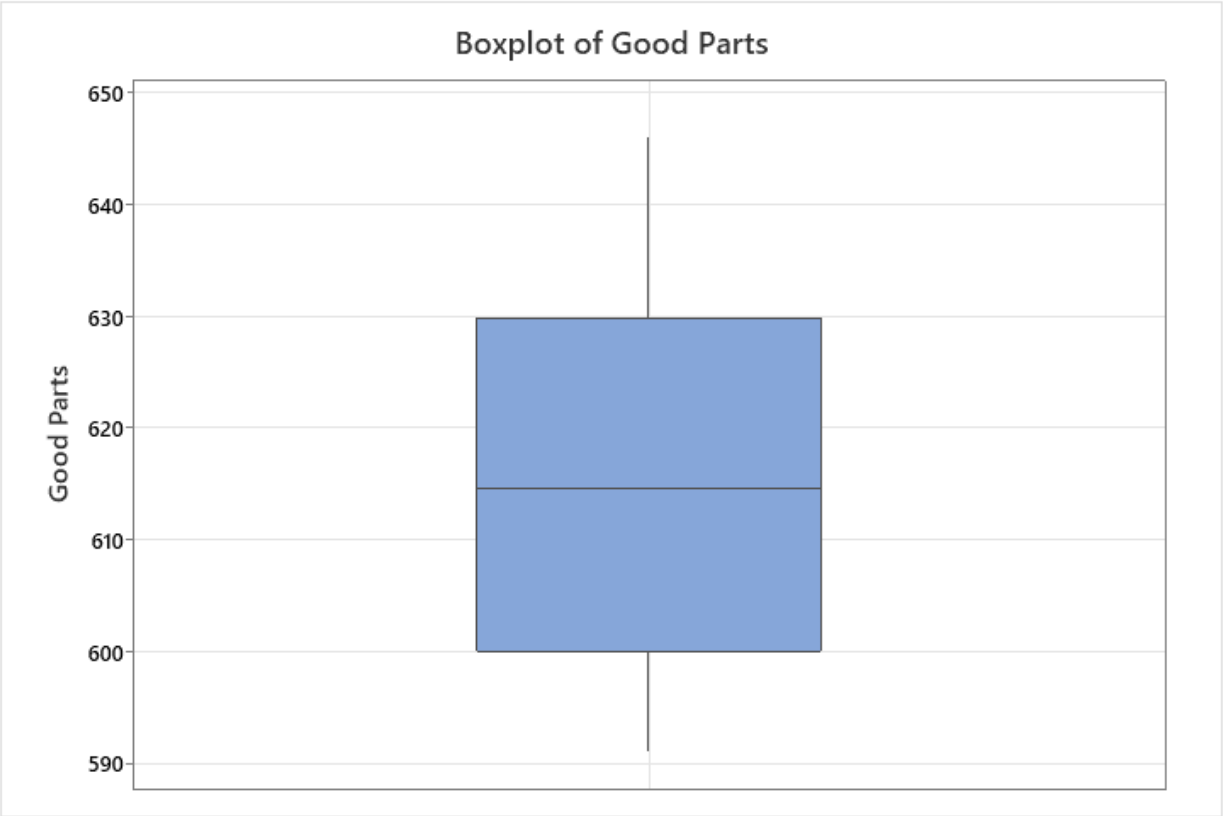

Figure 11. Box Plot for Phase 2 Good Parts

The distribution centers around roughly 610–615 good parts, with a relatively narrow interquartile range indicating consistent day-to-day output. The whiskers extend to reasonable limits with no outliers, suggesting that even the lowest and highest good-part counts remained within an expected range. Compared to the previous boxplot, the Phase 2 results show a slightly higher median and upper range, indicating an improvement in good-part production.

3.2.6. Off-spec Parts Numerical Comparison

Table 5 shows basic descriptive statistics for the Off-spec Parts produced according to Phase 2 data.

Table 5. Basic Statistics for Phase 2 Off-spec Parts

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Offspec Parts	20	0	61.7	2.03664	9.10812	46	55.25	61.5	69	79

The mean is 61.7, which is slightly smaller than the previous mean of 63.80. The standard deviation is 9.10812, which is slightly larger than the previous standard deviation of 8.21. The 95% confidence interval is (57.44, 65.96), which is similar to the previous interval of (59.96, 67.64).

A one-sample t-test produces a p-value of around 0.448, indicating that there is no statistical difference between the control chart data with the previous data.

3.2.7. Off-spec Parts Graphical Analysis and Comparison

A run chart of off-spec parts in Figure 12 was created to assess whether Phase 2 production exhibited any signs of nonrandom behavior.

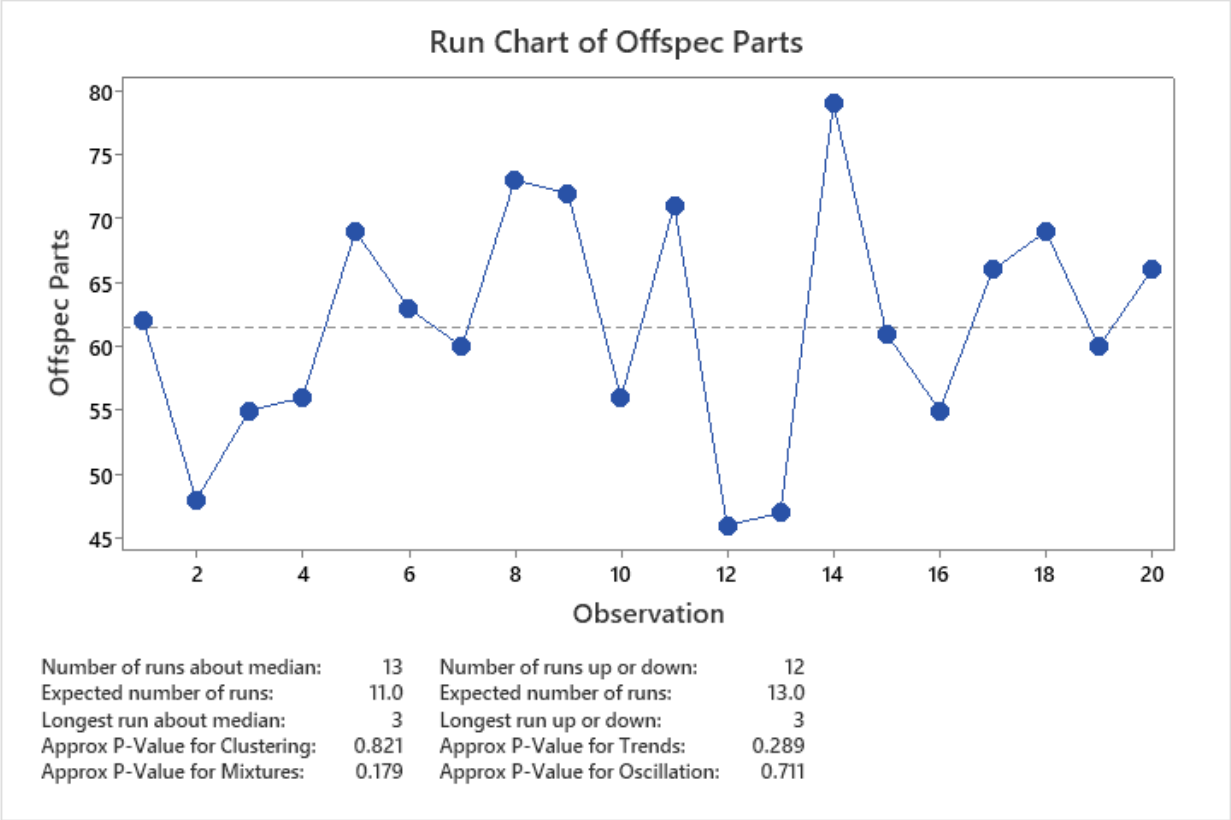

Figure 12. Run Chart for Phase 2 Off-spec Parts

The chart shows that off-spec counts fluctuate around the median in a pattern consistent with routine process variation, and the run test results support this interpretation: run counts closely match their expected values, and the p-values for clustering, mixtures, trends, and oscillation all indicate no significant nonrandom structure. This suggests that variation in off-spec output is stable and not driven by special causes. Compared to the previous run chart, the Phase 2 pattern appears similarly random and stable, with both charts showing no meaningful deviations from expected run behavior.

A histogram of daily off-spec parts in Figure 13 was created to examine how frequently the process produced units outside specifications during Phase 2.

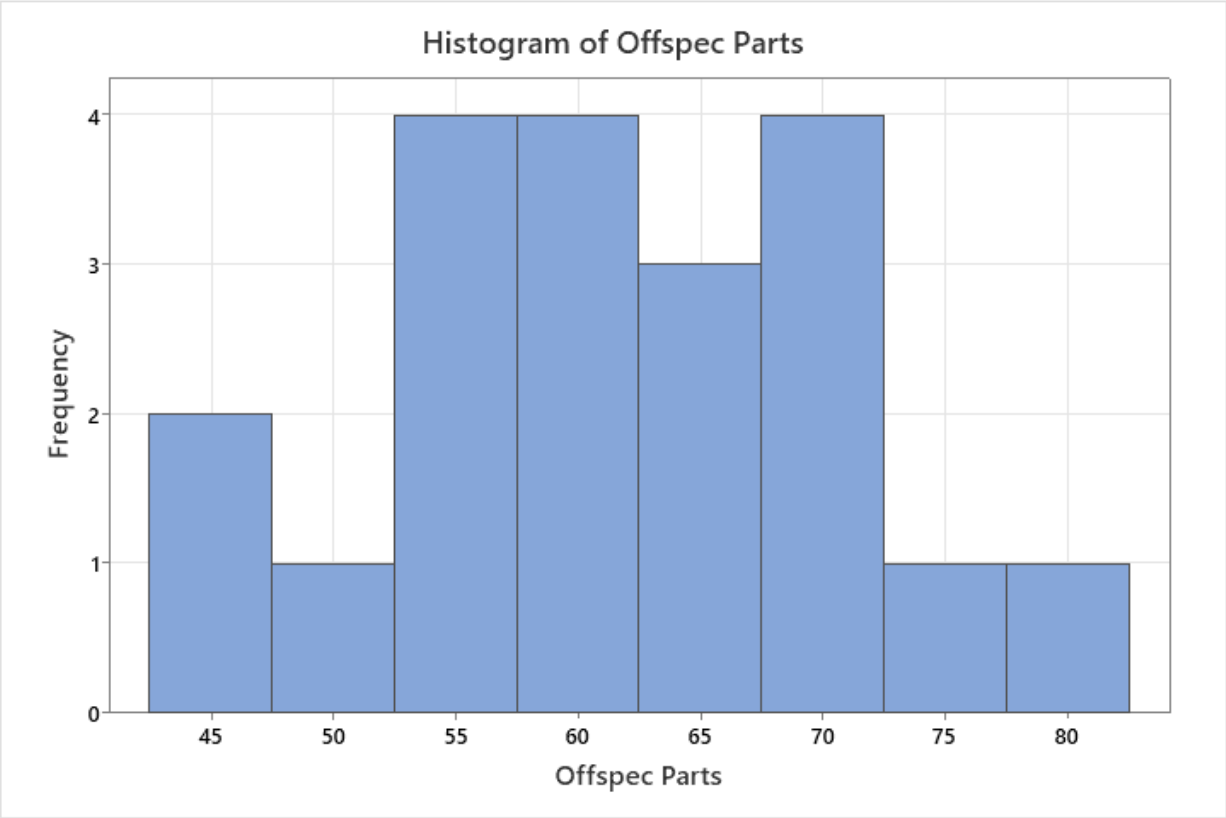

Figure 13. Histogram of Phase 2 Off-spec Parts

The distribution ranges from about 45 to 80 off-spec parts per day, with most observations clustered between 55 and 70, indicating moderate but steady variation without extreme values. Compared to the previous histogram, the Phase 2 results show a similar overall pattern, with off-spec counts concentrated in the same range and no meaningful shift in central tendency or spread.

A normal probability plot of daily off-spec parts in Figure 14 was generated to assess whether the variation in off-spec output follows a normal distribution during Phase 2.

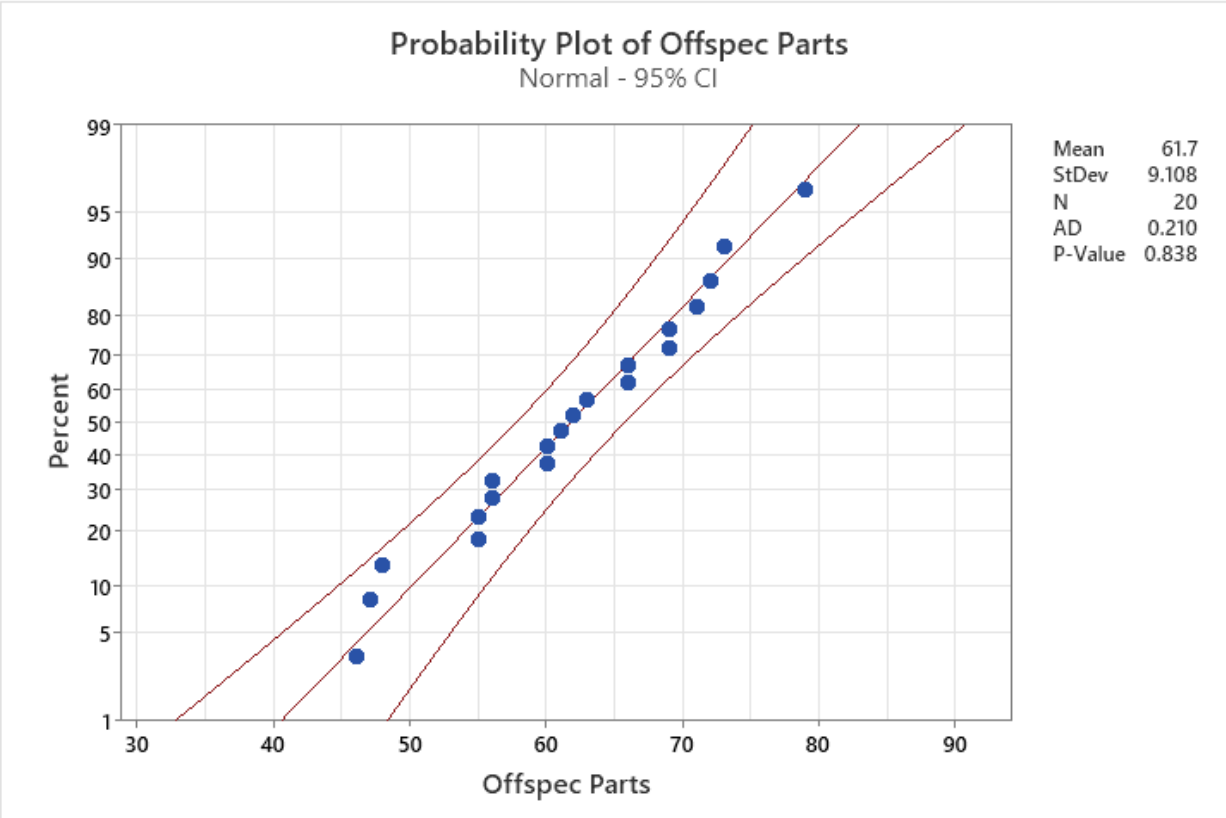

Figure 14. Probability Plot of Phase 2 Off-spec Parts

The points align closely with the reference line and remain well within the confidence bands, indicating no evidence of non-normal behavior. The high p-value (0.838) further supports a good fit to normality, suggesting that off-spec counts vary in a stable and predictable pattern suitable for statistical comparison. Compared to the previous probability plot, the Phase 2 results show a similarly strong alignment with the normal reference line, indicating no meaningful change in the distributional behavior of off-spec output.

A boxplot of daily off-spec parts in Figure 15 was created to summarize the central tendency and variability of out-of-spec production during Phase 2.

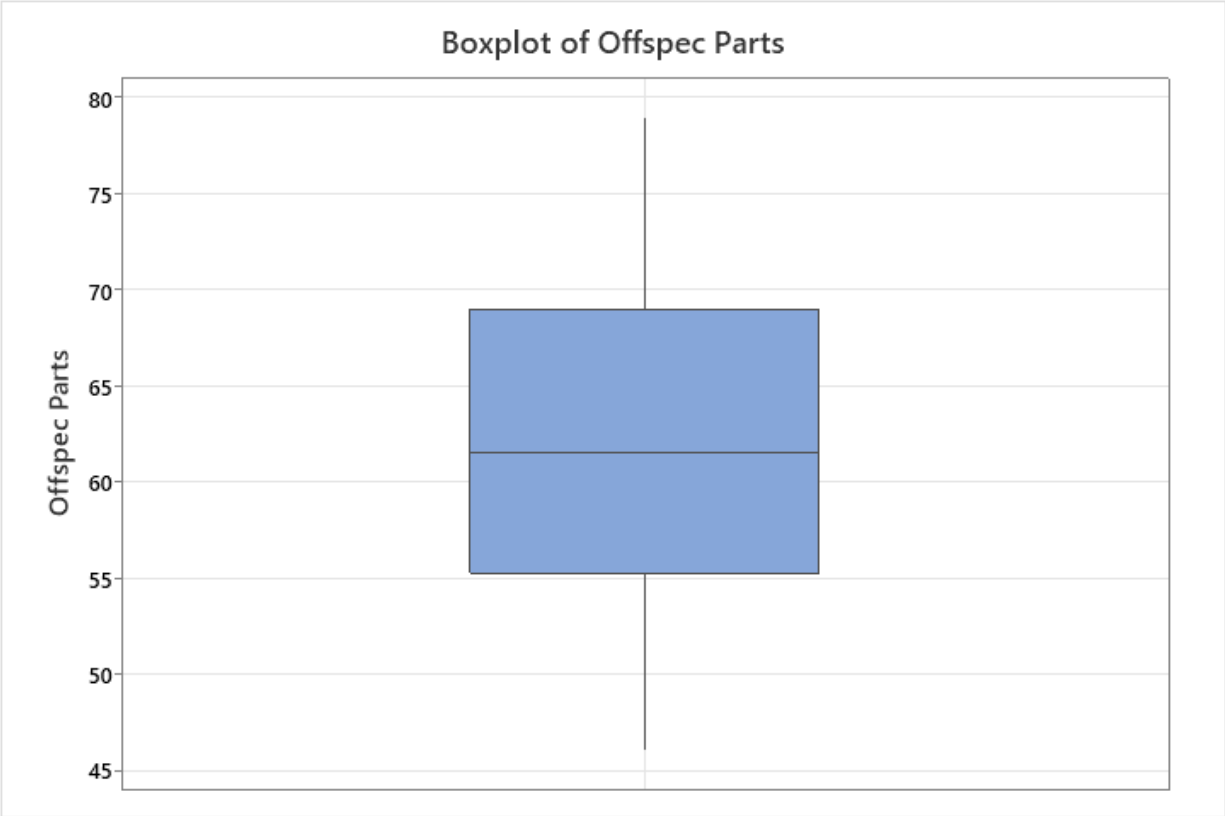

Figure 15. Box Plot of Phase 2 Off-spec Parts

The boxplot shows that off-spec output centers around roughly 60–62 units, with a moderate interquartile range indicating stable day-to-day variation. The whiskers extend to reasonable bounds without any extreme outliers, suggesting that the lowest and highest off-spec counts remained within an expected performance range. Compared to the previous boxplot, the Phase 2 results show a very similar median and spread, indicating that the overall level and variability of off-spec production remained largely unchanged.

3.2.8. Bad Parts Numerical Comparison

Table 6 shows basic descriptive statistics for the Bad Parts produced according to Phase 2 data.

Table 6. Basic Statistics for Phase 2 Bad Parts

Statistics

Variable	N	N*	Mean	SE Mean	StDev	Minimum	Q1	Median	Q3	Maximum
Bad Parts	20	0	123.6	2.88772	12.9143	102	111	125	130.25	152

The mean is 123.6, which is smaller than the previous mean of 151.20. The standard deviation is 12.9143, which is higher than the previous standard deviation of 7.62. The 95% confidence interval is (116.1, 131.1), which is wider but shifted down compared to the previous interval of (147.63, 154.77).

A one-sample t-test produces a p-value less than 0.0001, indicating that the difference is statistically significant.

3.2.9. Bad Parts Graphical Analysis and Comparison

A run chart of bad parts in Figure 16 was constructed to evaluate whether Phase 2 production displayed any signs of nonrandom or unstable behavior.

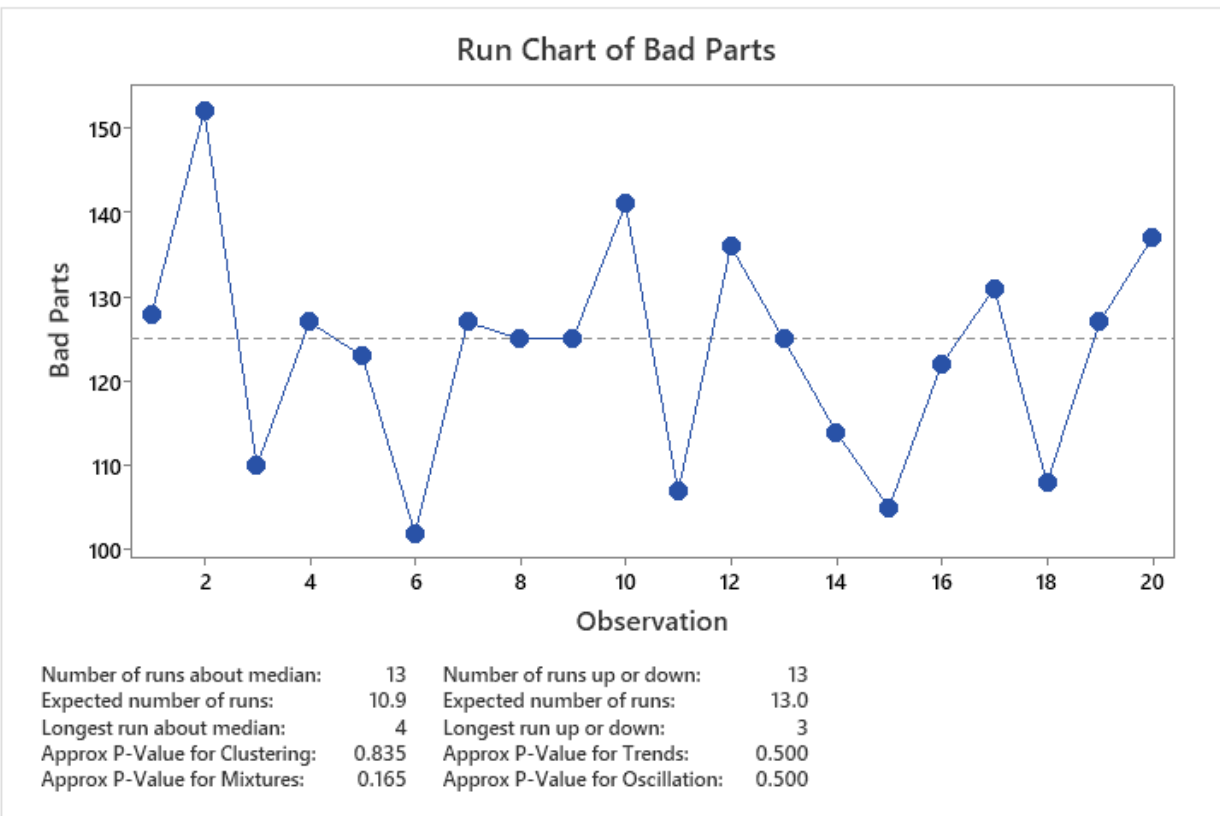

Figure 16. Run Chart of Phase 2 Bad Parts

The chart shows that bad-part counts fluctuate around the median with no clear directional trends, and the run test statistics indicate no evidence of clustering, mixtures, trends, or oscillation. Although the number of runs about the median is slightly lower than expected, the p-values remain well above typical significance thresholds, suggesting the variation is still consistent with routine process noise. Compared to the previous run chart, Phase 2 displays a similar overall pattern of random fluctuation, with slightly longer sequences on one side of the median but no indications of meaningful nonrandom behavior in either period.

A histogram of daily bad parts in Figure 17 was created to evaluate how frequently the process produced defective units during Phase 2.

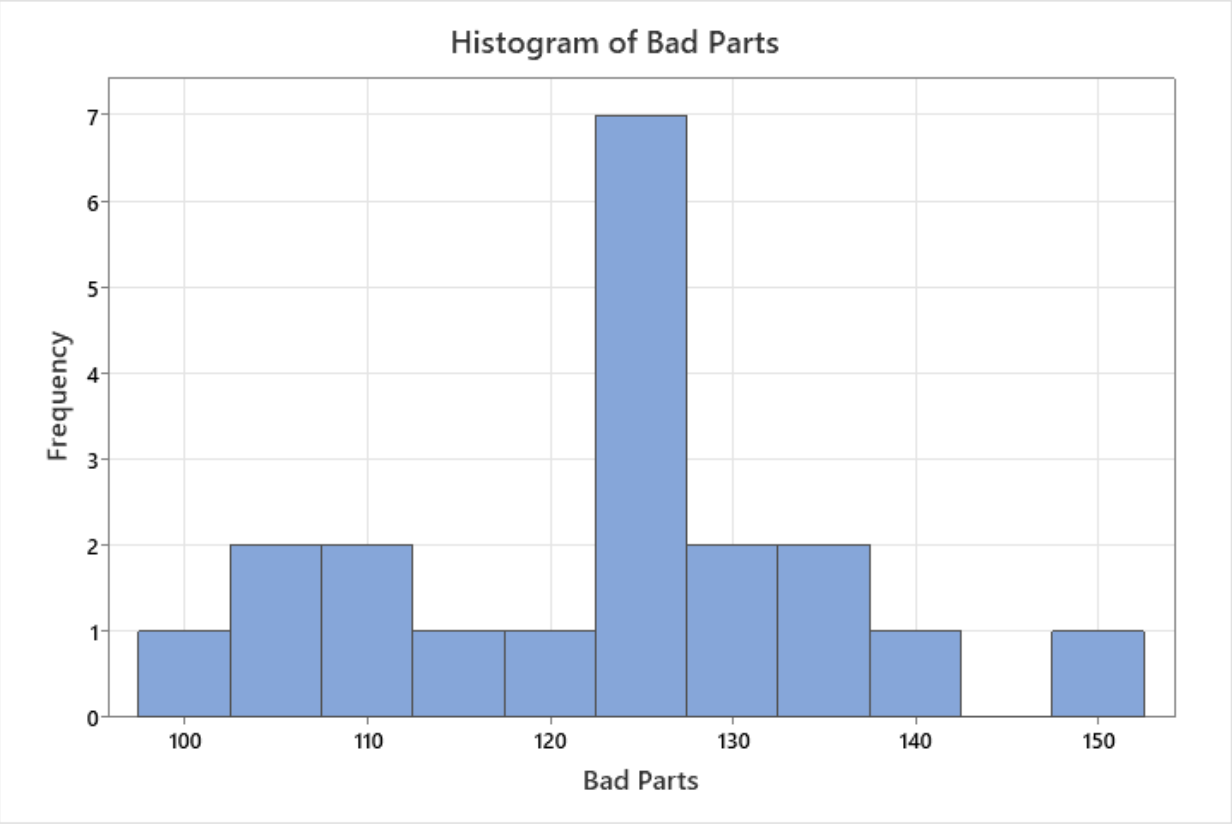

Figure 17. Histogram of Phase 2 Bad Parts

The distribution of bad parts ranges from roughly 95 to 150 units per day, with most observations clustering around 120–130 units, indicating a generally consistent level of defective output with some day-to-day variation. No extreme spikes appear, suggesting that bad-part production stayed within a typical operating range. Compared to the previous histogram, the Phase 2 distribution shows a noticeably lower center and a wider spread,

indicating that the process produced fewer bad parts on average but with greater variability across days.

A normal probability plot of daily bad parts in Figure 18 was generated to evaluate whether defective output follows a normal distribution during Phase 2.

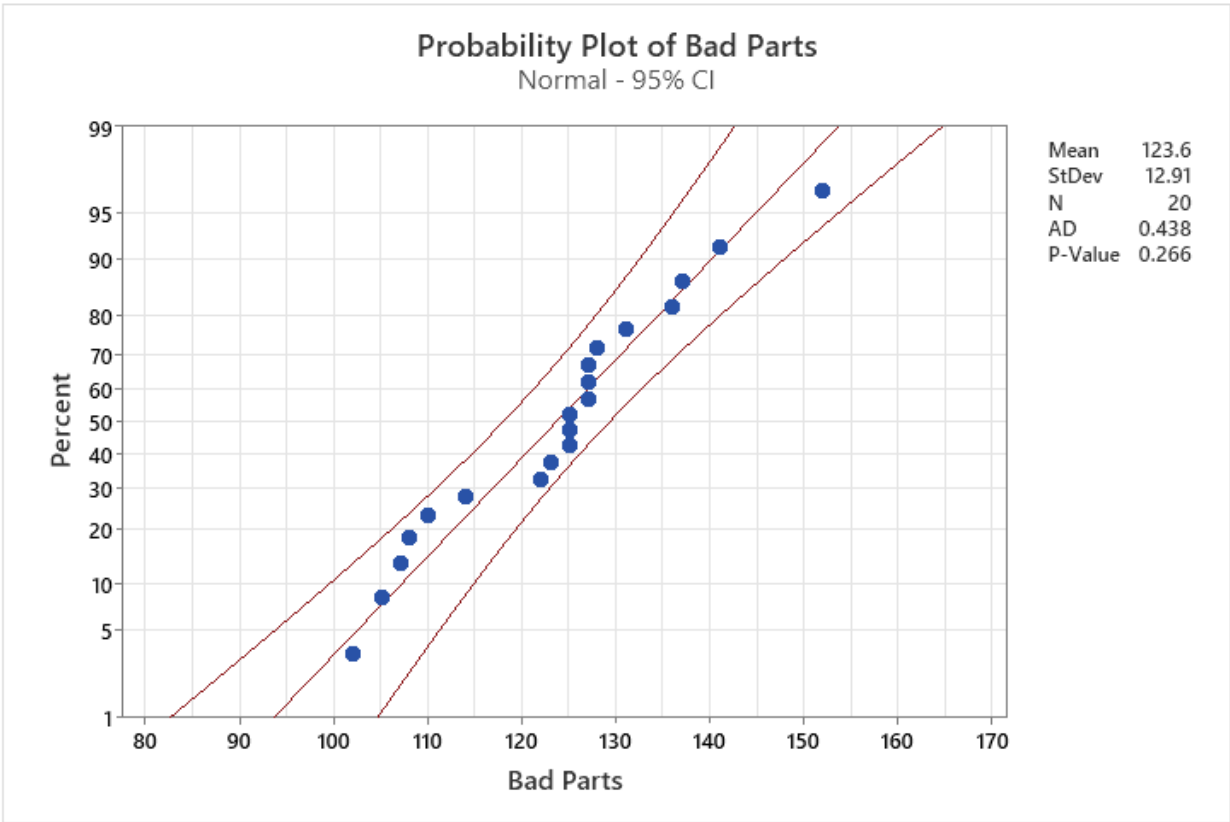

Figure 18. Probability Plot of Phase 2 Bad Parts

The plotted points follow the reference line closely and remain within the confidence bands, indicating no meaningful departures from normality. The p-value of 0.266 provides additional support that the distribution of bad-part counts is consistent with a normal model and varies in a stable, predictable pattern. Compared to the previous probability plot, the Phase 2 results show a similarly strong alignment with the normal reference line, indicating that the distributional behavior of bad-part output remained largely unchanged across periods.

A boxplot of daily bad parts in Figure 19 was created to summarize the central tendency and variability of defective output in Phase 2.

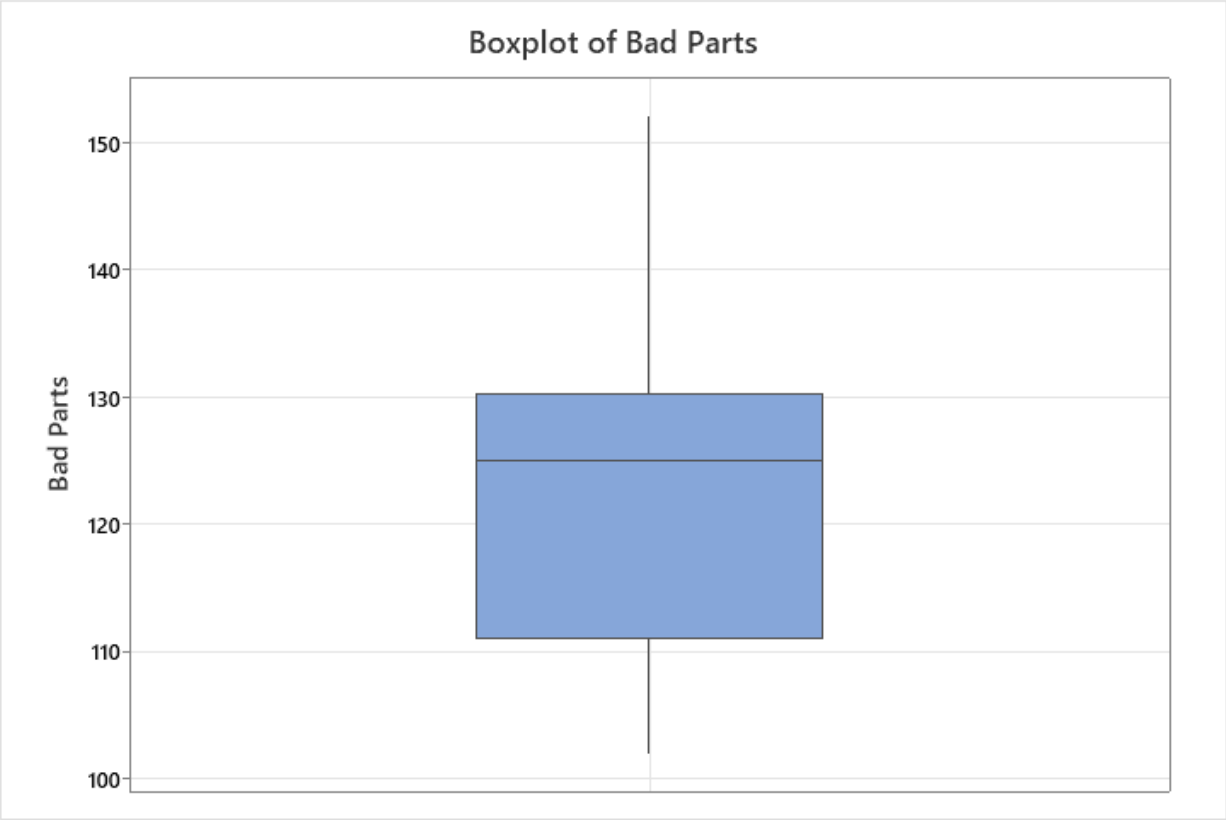

Figure 19. Box Plot of Phase 2 Bad Parts

The boxplot indicates that bad parts cluster around a median of roughly 125, with most observations falling between about 112 and 130, showing moderate consistency and no extreme outliers. The whiskers extend from just over 100 to approximately 152, suggesting only occasional higher-defect days. Compared to the previous boxplot, Phase 2 shows a clear downward shift in both the median and overall spread of bad parts, indicating that implementing the control-chart-based process helped reduce defect counts and improve production quality.

4. Conclusions

The Phase 2 production data shows that the implementation of statistical process control produced a mixed set of outcomes when compared to the baseline results from Lab 5. Good part output increased meaningfully, and the corresponding t test confirmed that this difference was statistically significant with a p value below 0.001. The confidence interval for the Phase 2 mean was also shifted upward relative to the interval from Lab 5, indicating that the improvement in conforming units is unlikely to be due to random chance. Bad part production decreased significantly as well, with its t test yielding a p value below 0.0001.

The confidence interval for the mean number of bad parts also moved downward, supporting the conclusion that Phase 2 achieved a measurable reduction in defects.

Off-spec parts and total parts did not show statistically significant changes. Their p-values were both above 0.40, and the confidence intervals overlapped with those from Lab 5, which indicates that any apparent shifts in these categories are consistent with random variation. Run charts for all variables displayed stable behavior with no special cause signals, and the probability plots showed that the underlying distributions remained consistent with normality. These results together suggest that the primary improvement produced by the control chart was related to quality rather than output volume. More good parts were produced, and fewer bad parts were generated, but the overall production level and intermediate off-spec counts did not undergo meaningful change.

5. Recommendations

The significant improvement in good part production and the substantial reduction in bad parts indicate that the control chart should continue to be used as part of regular process monitoring. The control chart has demonstrated its ability to identify and support the correction of unwanted variation, and it provides an effective framework for improving quality outcomes. However, further work is necessary to address the lack of change in total output and off-spec counts. These areas point to underlying process behaviors that were not fully resolved by the introduction of the control chart.

A focused investigation should be conducted into the persistent shift in Button-Hole Diameter, particularly because Phase 2 time series plots showed repeated observations exceeding the upper control limit. The process should also incorporate a more responsive operator feedback mechanism so that out-of-control conditions trigger timely corrective action rather than being passively recorded. Additional process variables should be monitored with their own control charts so that more potential sources of variation are captured and managed. Increasing the sampling frequency or adjusting subgroup construction may also improve the sensitivity of monitoring and help detect systematic changes earlier. These steps would help extend the quality gains observed in Phase 2 and support continued reduction of variability.